1.2.3 Making Lists of Objects

In doing calculations, it is often convenient to collect together several objects, and treat them as a single entity. *Lists* give you a way to make collections of objects in *Mathematica*.

A list like `{3, 1, 6, 2}` is a collection of four objects. You can treat it just like a single object. You can do arithmetic with it, assign it as the value of a variable, and so on.

Here is a list of three numbers.

```
In[1]:= {3, 5, 1}
Out[1]= {3, 5, 1}
```

This adds 2 to each of the numbers in the list.

```
In[2]:= % + 2
Out[2]= {5, 7, 3}
```

You can do any arithmetic operation on the list.

```
In[3]:= %^2 + % + 1
Out[3]= {31, 57, 13}
```

You can name the list `v`.

```
In[4]:= v = %
Out[4]= {31, 57, 13}
```

When you ask for `v`, you get the list.

```
In[5]:= v
Out[5]= {31, 57, 13}
```

Now you can subtract one from each element of `v`.

```
In[6]:= v - 1
Out[6]= {30, 56, 12}
```

Like other arithmetic operations, division is done on each element of the lists in turn.

```
In[7]:= v / (v - 1)
```

Out[7]= $\{\frac{31}{30}, \frac{57}{56}, \frac{13}{12}\}$

All the mathematical functions in Section 1.1.3 can be applied to lists.

```
In[8]:= Exp[-2 %] / 5  //N
Out[8]= {0.0253214, 0.0261174, 0.0229118}
```

`Range[`n`]`	create the list `{1, 2, 3, ..., `n`}`
`Range[`n_1`, `n_2`]`	create the list `{`n_1`, `n_1`+1, ..., `n_2`}`
`Range[`n_1`, `n_2`, `dn`]`	create the list `{`n_1`, `n_1`+`dn`, ..., `n_2`}`

A functions that creates lists.

This gives a list of the first five integers.

```
In[9]:= Range[5]
Out[9]= {1, 2, 3, 4, 5}
```

Here are the first five squares.

```
In[10]:= Range[5]^2
Out[10]= {1, 4, 9, 16, 25}
```

This gives a list of integers in the range -2 to 2.

```
In[11]:= Range[-2, 2]
Out[11]= {-2, -1, 0, 1, 2}
```

This gives a list of numbers from 0 to 1 in steps of 0.2.

```
In[12]:= Range[0, 1, 0.2]
Out[12]= {0, 0.2, 0.4, 0.6, 0.8, 1.}
```

Section 1.7 discusses many other ways to create lists in *Mathematica*.